

16 PBRs

PBR mole balance:

- Let's review PFRs by doing an example: Gas phase reaction $A + 0.5 B \rightarrow C$, carried out in an isothermal, isobaric PFR. Determine the V needed for 60% conversion. The feed is stoichiometric in A & B. Additionally:

$$\circ -r_A = k P_A^{\frac{1}{3}} P_B^{\frac{2}{3}}, k = 0.00392 \text{ mol/L} \cdot \text{atm} \cdot \text{s}, F_{A0} = 0.1362 \text{ mol/s}, P_0 = 10 \text{ atm}$$

PFR mole balance:

$$F_{A0} \frac{dX_A}{dV} = -r_A$$

Rate law:

$$-r_A = k P_A^{\frac{1}{3}} P_B^{\frac{2}{3}}$$

Expressions for P_i : Use Eq 4-27: $P_j = \frac{P_{A0}(\Theta_j + \nu_j X_A)}{1 + \epsilon X_A}$

$$P_A = \frac{P_{A0}(1 - X_A)}{1 + \epsilon X_A}, \quad P_B = \frac{P_{A0}(\Theta_B - \frac{1}{2} X_A)}{1 + \epsilon X_A}$$

$$\Theta_B = \frac{F_{B0}}{F_{A0}} = \frac{1}{2}$$

δ and ϵ :

$$\delta = -\frac{1}{2}, \quad \epsilon = Y_{A0} \delta = -\frac{1}{3}$$

$$Y_{A0} = \frac{1 \text{ mol A}}{1 \text{ mol A} + \frac{1}{2} \text{ mol B}} = \frac{2}{3}$$

Limits of integration:

X goes from 0 to 0.6, V goes from 0 to V

Solution: $V = 6.71$ L. You should verify that you can get this solution. Try it!

- Now, let's compare and contrast PFRs and PBRs:

Similarities								Differences
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tubular reactors

reaction proceeds down
the length (or volume)
of the reactor

plug flow through the
reactor $\rightarrow$ turbulent
flow

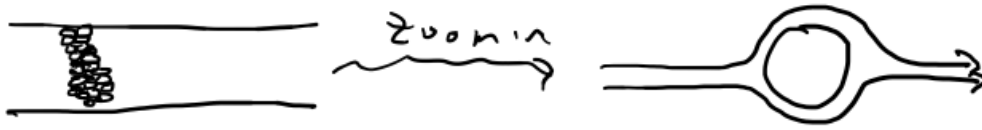
perfect mixing in the radial
dimension

continuous variation in the
axial dimension (down
the length)

PBR is packed with
catalyst pellets

gases in the PBR may
experience pressure
drop due to the
packing

- What is meant by "packed bed"?



- flow around a spherical particle

- Note: In this class, it's OK to think about packing as spherical pellets.

- Flow around the particle causes drag

- Drag causes pressure drop

- Decreased pressure causes concentrations to decrease

- Decreased concentration influences the rate law

- Account for pressure drop using Ergun Equation

- Ergun Equation (full expression): $\frac{dP}{dz} = -\frac{G}{\rho g_c D_p} \left(\frac{1-\phi}{\phi^2} \right) \left[\frac{150(1-\phi)\mu}{D_p} + 1.75G \right]$

- In this equation:

$\frac{150(1-\phi)\mu}{D_p}$ dominates in laminar flow
 $1.75G$ dominates in turbulent flow

- Important terms:

P (capital P) = total pressure

z = reactor length = $\frac{\text{volume}}{\text{cross sectional area}}$

D_p = catalyst pellet diameter

- Ergun Equation, simplified forms that are useful for CRE/CRD (Equations 5-28 and 5-30 in Fogler):

$$\frac{dp}{dW} = -\frac{\alpha}{2p} \frac{T}{T_0} \frac{F_T}{F_{T0}} \quad \text{or} \quad \frac{dp}{dW} = -\frac{\alpha}{2p} \frac{T}{T_0} (1 + \varepsilon X)$$

where p (lower case) = $\frac{P}{P_0}$ ← total pressure at some location within the PBR

α = pressure drop parameter, has units of inverse mass
 P_0 ← total pressure @ the reactor inlet

- Hence, relationship between D_p and α : $\uparrow D_p \downarrow \alpha$, $\downarrow D_p \uparrow \alpha$

smaller catalyst pellets $\rightarrow$ more pressure drop
 however larger catalyst pellets mean larger paths for diffusion of molecules in the catalyst pores, $\therefore$ this can impede the reaction rate,

- Application:

Ergun Equation couples with the PBR mole balance

- Solving this set of coupled ODEs usually requires a numerical solution using software. EXCEPTION: $\epsilon = 0$ and $\frac{T}{T_0} = 1$. This can be solved analytically.
- Let's repeat the example from the beginning of the lecture, this time it's a PBR with pressure drop, i.e., it's not isobaric. Other changes:
 - $k = 0.00392 \text{ mol/atm} \cdot \text{kgcat} \cdot \text{s}$, $\alpha = 0.0367 \text{ kg}^{-1}$

Mole balance:
$$F_{A0} \frac{dX_A}{dW} = -r_A$$

What else is different?

① need to include Ergun Eq

$$\frac{dp}{dW} = - \frac{\alpha}{Z_p} \left(1 - \frac{1}{3} X_A \right)$$

Limits of integration: p goes from 1 to p

W goes from 0 to W

② expressions for partial pressures $\rightarrow$ Eq 4-27.

$$P_A = \frac{P_{A0}(1-X_A)}{1 - \frac{1}{3} X_A} P$$

$$P_B = \frac{\frac{1}{2} P_{A0}(1-X_A)}{1 - \frac{1}{3} X_A} P$$

lower case p , takes into account the effect of pressure drop on volumetric flowrate.

Key point: pressure drop influences volumetric flowrate $\frac{1}{2}$ hence concentration $\frac{1}{2}$, this is true even in cases where $\delta = 0$.